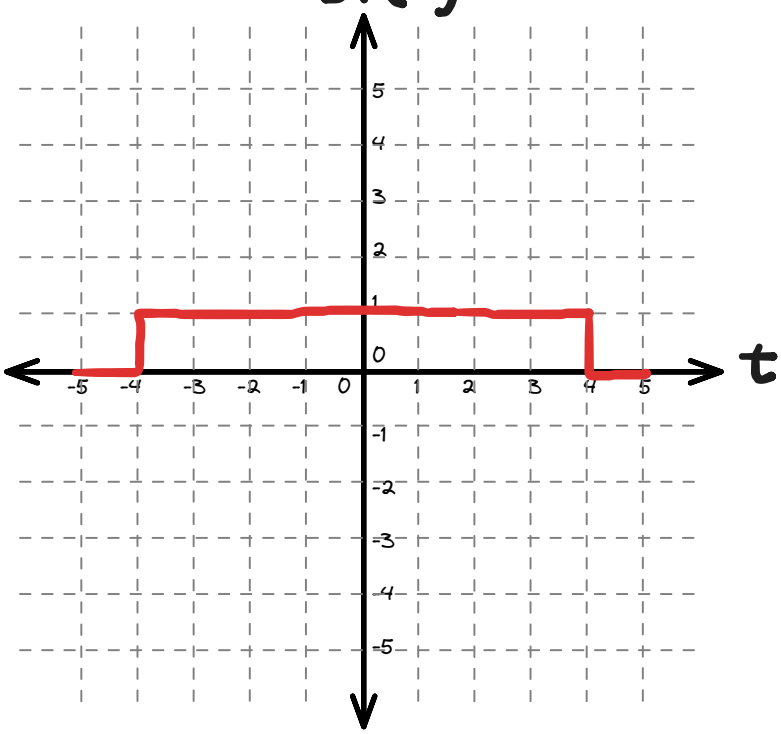
Last Year Exam

Question 1 (CH 1)

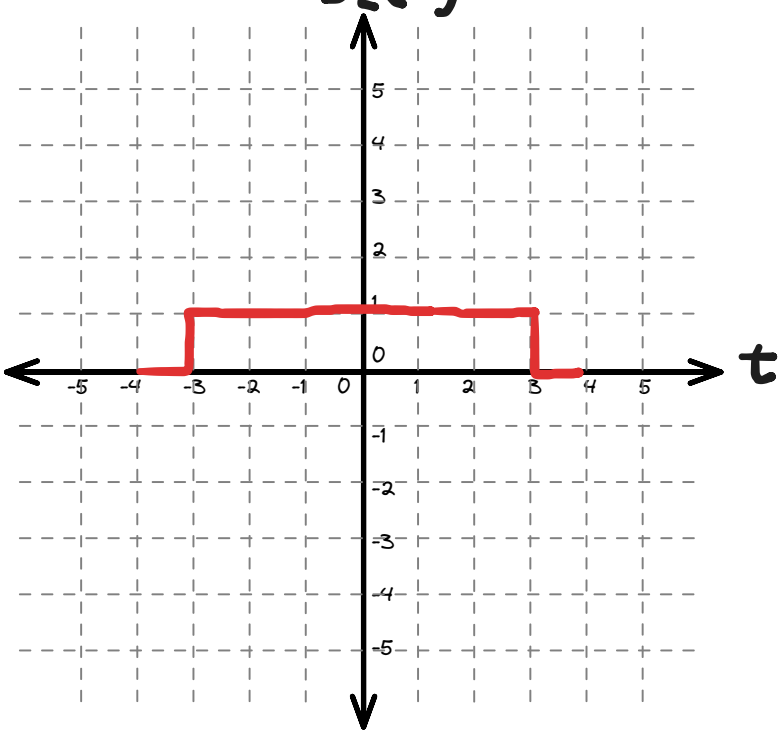
(a)

$g_1(t) = g\left(\frac{t}{4}\right)$
$g_2(t) = g\left(\frac{t}{3}\right)$
$g_3(t) = g(t)$
$x(t) = g_1(t) + g_2(t) + g_3(t)$

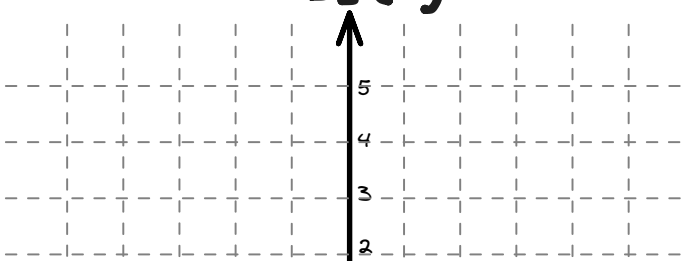
$g_1(t)$

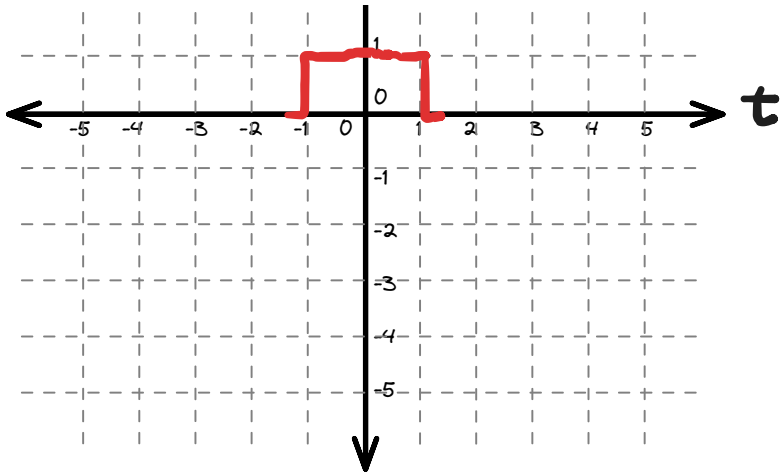

t
-5 -4 -3 -2 -1 0 1 2 3 4 5
5 4 3 2 1 0 -1 -2 -3 -4 -5

$g_2(t)$


t
-5 -4 -3 -2 -1 0 1 2 3 4 5
5 4 3 2 1 0 -1 -2 -3 -4 -5

$g_3(t)$


5 4 3 2



(b)

(i)

Since it is a continuous sinusoid function, it is periodic. The fundamental period can be calculated with the fundamental frequency of 2π in this case

$$T_0 = \frac{2\pi}{2\pi} = 1$$

(ii)

$$\begin{aligned} \frac{7}{15}\pi N &= 2\pi m, \quad m \in \mathbb{Z} \\ \frac{7}{15} N &= 2m \\ N &= \frac{30}{7} m, \quad m = 7 \\ N &= 30 \end{aligned}$$

(iii)

no

(c)

It is **not memoryless** since it depends on the future value at time t^2.

It is **not causal** since it depends on the future value at time t^2.

To check time invariance the time invariance property is checked, so looking at an input

$$y_1(t) = x_1(t^2)$$

The corresponding input shifted in time would be

$$x_2(t^2) = x_1(t^2 - t_0)$$

Shifting the output would result in

$$y_2(t) = y_1(t - t_0) = x_1((t - t_0)^2)$$

Since

$$x_1(t^2 - t_0) \neq x_1((t - t_0)^2)$$

the system is not time invariant.

To check linearity we look at two inputs $x_1(t)$ and $x_2(t)$, where a linear combination combination $x_3(t) = ax_1(t) + bx_2(t)$ must give a similarly weighted linear combination of the corresponding outputs.

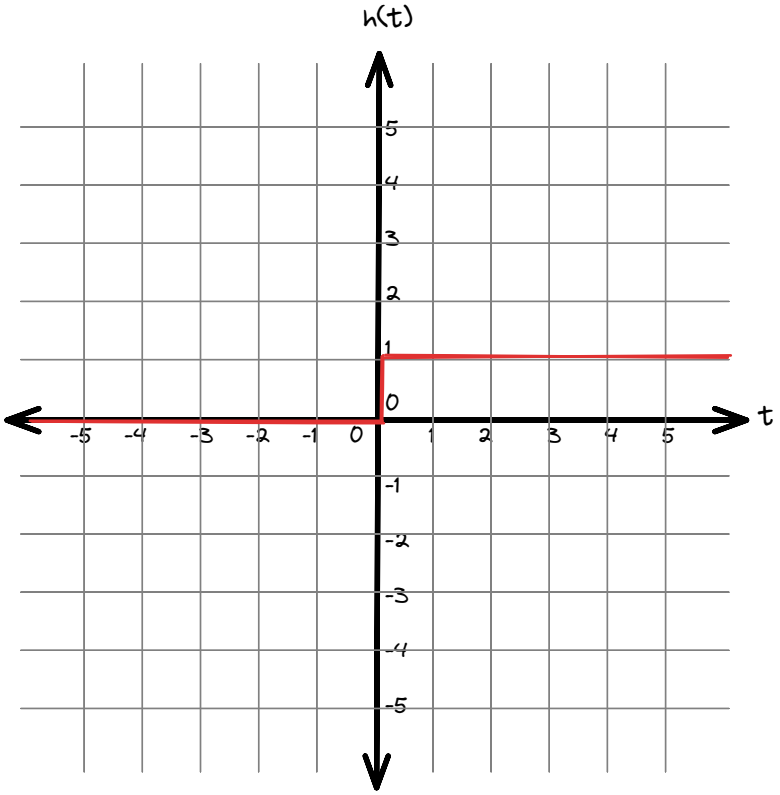
$$\begin{aligned} y_3(t) &= x_3(t^2) \\ &= ax_1(t^2) + bx_2(t^2) \\ &= ay_1(t) + by_2(t) \end{aligned}$$

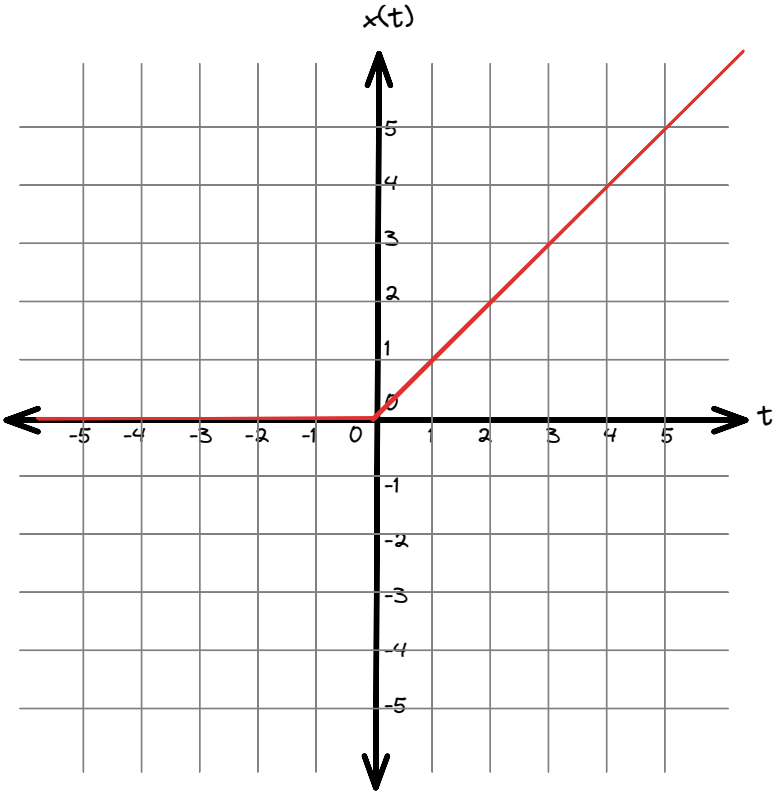
Since the linear combinations match up **it is a linear system**.

Question 2 (CH 2)

(a)

(i)

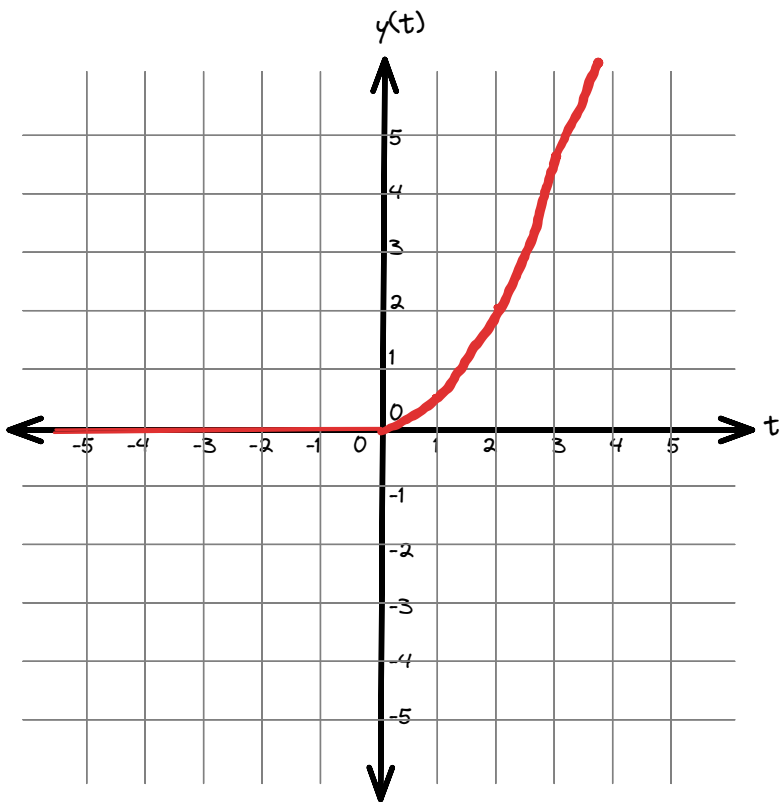

h(t)
t
5
4
3
2
1
0
-1
-2
-3
-4
-5
-5 -4 -3 -2 -1 0 1 2 3 4 5


x(t)
t
5
4
3
2
1
0
-1
-2
-3
-4
-5
-5 -4 -3 -2 -1 0 1 2 3 4 5

(ii)

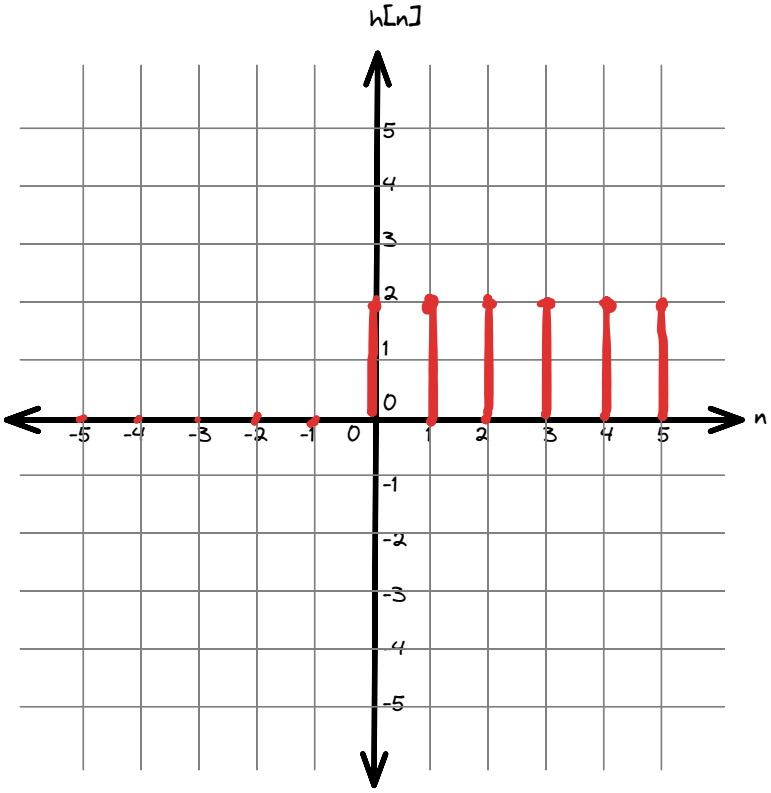
$$
\begin{aligned}
y(t) &= \int_{-\infty}^{\infty} \tau u(\tau) u(t-\tau)\, d\tau \\
&= \int_{0}^{t} \tau\, d\tau \\
&= \frac{1}{2}\left[\tau^2\right]_0^t \\
&= \frac{1}{2}(t^2 - 0^2) \\
&= \frac{t^2}{2}
\end{aligned}
$$

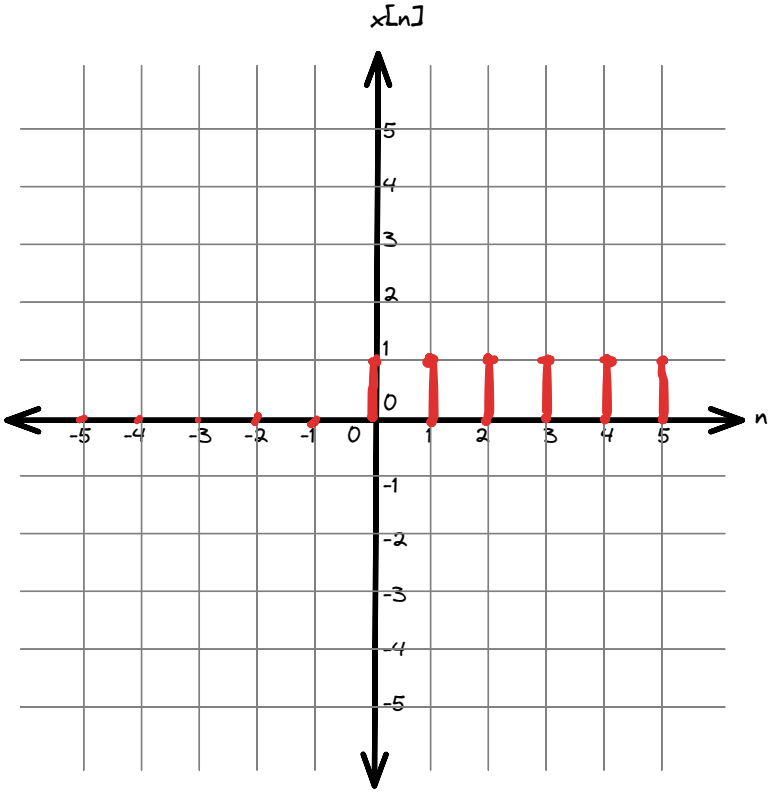
(iii)



(b)

(i)

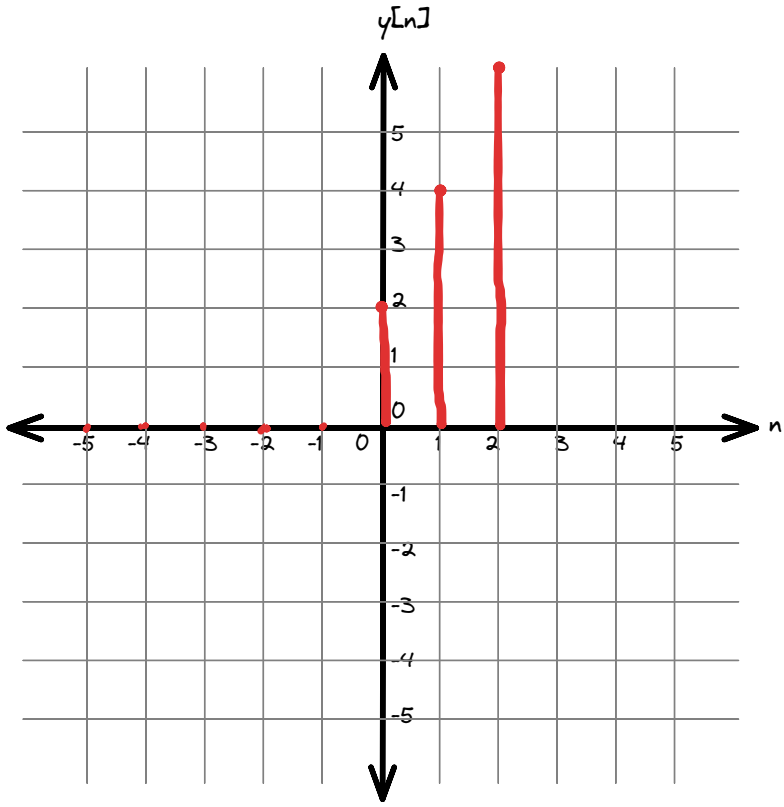

h[n]
n
5
4
3
2
1
0
-1
-2
-3
-4
-5
-5 -4 -3 -2 -1 0 1 2 3 4 5


x[n]
n
5
4
3
2
1
0
-1
-2
-3
-4
-5
-5 -4 -3 -2 -1 0 1 2 3 4 5

(ii)

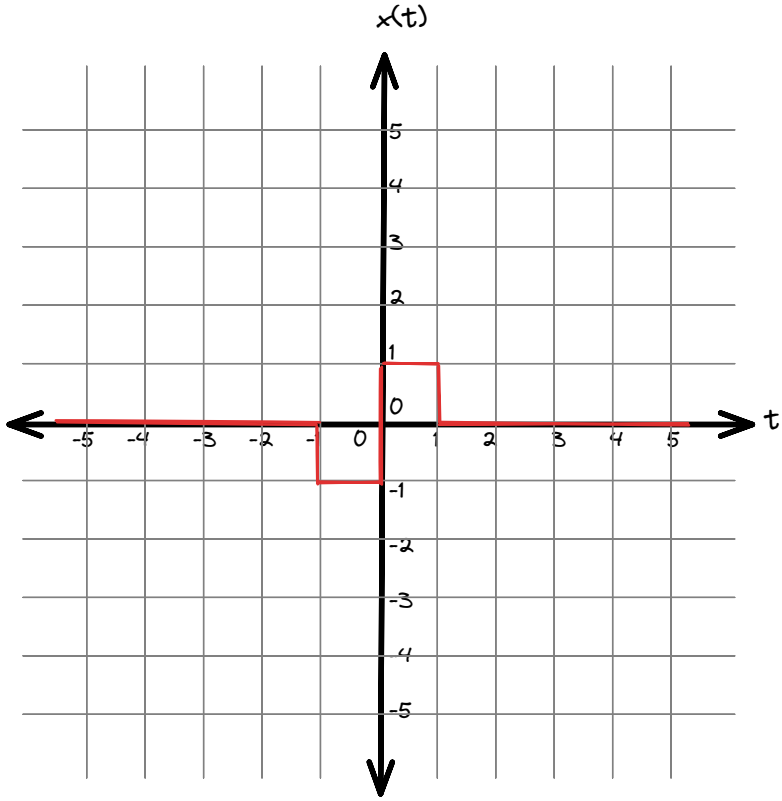
$$
\begin{aligned}
y[n] &= \sum_{k=-\infty}^{\infty} u[k]2u[n-k] \\
&= \sum_{k=0}^{n} 2 \\
&= \begin{cases} 2n+2, & n \geq 0 \\ 0, & n < 0 \end{cases}
\end{aligned}
$$

(iii)



Question 3 (CH 3 + 4)

(i)



(ii)

$$
\begin{aligned}
X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt \\
&= \int_{-1}^{0} -1e^{-j\omega t}\,dt + \int_{0}^{1} 1e^{-j\omega t}\,dt \\
&= \frac{1}{j\omega}[e^{-j\omega t}]_{-1}^{0} - \frac{1}{j\omega}[e^{-j\omega t}]_{0}^{1}, \qquad \omega \neq 0 \\
&= \frac{1}{j\omega}(e^{0} - e^{j\omega}) - \frac{1}{j\omega}(e^{-j\omega} - e^{0}) \\
&= \frac{(1 - e^{j\omega}) - (e^{-j\omega} - 1)}{j\omega} \\
&= \frac{1 - e^{j\omega} - e^{-j\omega} + 1}{j\omega} \\
&= \frac{2 - \cos(\omega) - j\sin(\omega) - \cos(\omega) + j\sin(\omega)}{j\omega} \\
&= \frac{2 - 2\cos(\omega)}{j\omega} \\
&= \frac{4}{j\omega}\frac{1 - \cos(\omega)}{2} \\
&= \frac{4\sin^2\left(\frac{\omega}{2}\right)}{j\omega}
\end{aligned}
$$

Here ω can't be 0, so the final transform would be

$$X(j\omega) = \begin{cases} \frac{4\sin^2\left(\frac{\omega}{2}\right)}{j\omega}, & \omega \neq 0 \\ 0, & \omega = 0 \end{cases}$$

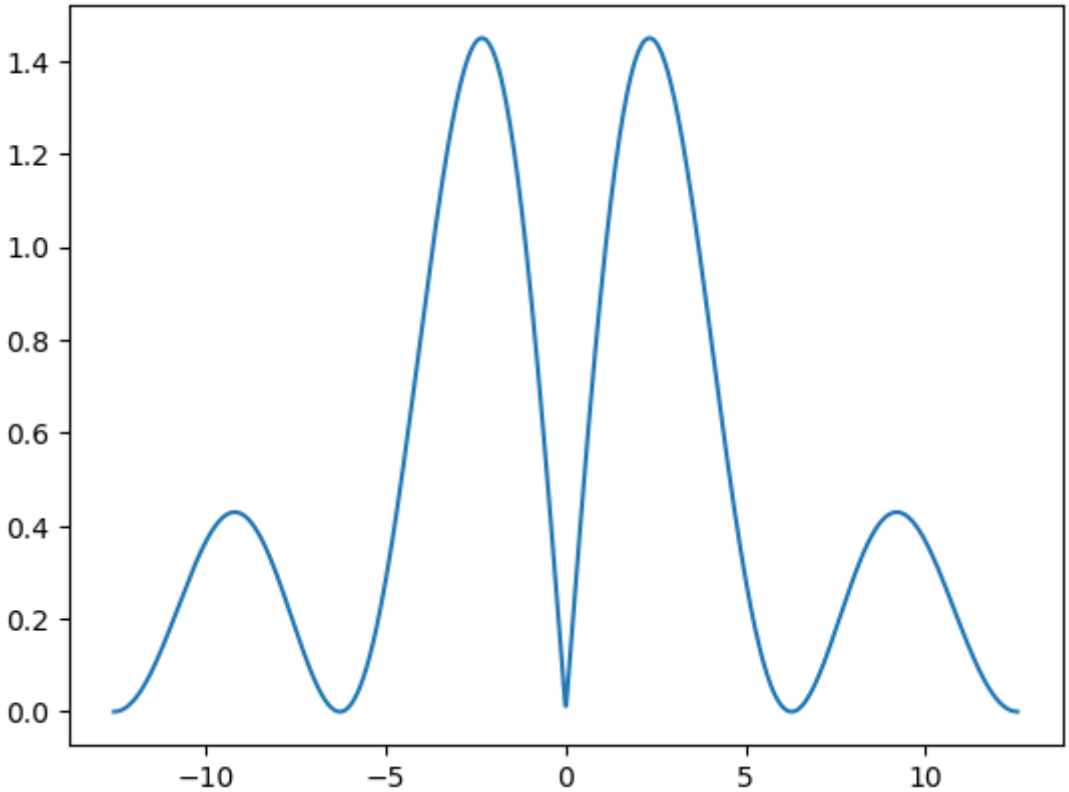
(iii)

$$\begin{aligned} |X(j\omega)| &= \left| \frac{4\sin^2\left(\frac{\omega}{2}\right)}{j\omega} \right| \\ &= \frac{|4\sin^2\left(\frac{\omega}{2}\right)|}{|j\omega|} \\ &= \frac{\sqrt{(4\sin^2\left(\frac{\omega}{2}\right))^2}}{\sqrt{\omega^2}} \\ &= \frac{|4\sin^2\left(\frac{\omega}{2}\right)|}{|\omega|} \\ &= 4\left| \frac{\sin^2\left(\frac{\omega}{2}\right)}{\omega} \right| \end{aligned}$$

(iv)

This function is too hard to sketch by hand so it is done in python

```
fig, ax = plt.subplots()
omega = np.linspace(-4*np.pi, 4*np.pi, 1000)
x = 4 * np.abs((np.sin(omega/2)**2) / omega)
ax.plot(omega, x)
```



Question 4 (CH 4 + 5 + 6)

(a)

(i)

Using equation 4.76 from page 331.

$$H(j\omega) = \frac{1}{5j\omega + 10}$$

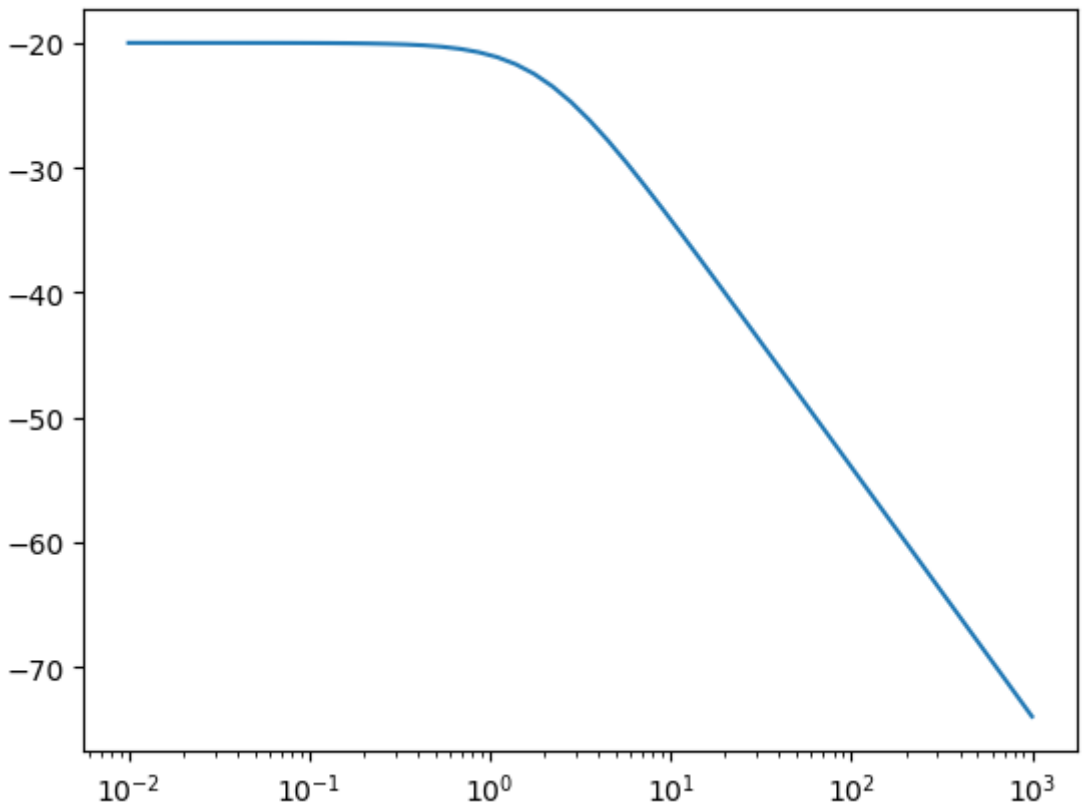
(ii)

Log magnitude is found

$$\begin{aligned} 20\log(|H(j\omega)|) &= 20\log\left(\left|\frac{1}{5j\omega + 10}\right|\right) \\ &= 20\log\left(\frac{|1|}{|5j\omega + 10|}\right) \\ &= 20\log\left(\frac{1}{\sqrt{100 + 25\omega^2}}\right) \end{aligned}$$

The bode plot is made in python

```
fig, ax = plt.subplots()
omega = np.logspace(-2, 3)
H = 20 * np.log10(1/np.sqrt(100 + 25 * omega**2))
ax.plot(omega, H)
plt.semilogx()
```



(b)

(i)

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$$
\begin{aligned}
H(e^{j\Omega}) &= \frac{1 + \frac{1}{2}e^{-j\Omega}}{1 - \frac{1}{2}e^{-j\Omega}} \\
&= \frac{2 + e^{-j\Omega}}{2 - e^{-j\Omega}}
\end{aligned}
$$

(ii)

The frequency response can be found for a complex number of the form $z = a + e^b$
The frequency response is rewritten to split the real and imaginary parts

$$
\begin{aligned}
H(e^{j\Omega}) &= \frac{2+e^{-j\Omega}}{2-e^{-j\Omega}} \\
&= \frac{2+\cos(\Omega)-j\sin(\Omega)}{2-\cos(\Omega)+j\sin(\Omega)} \\
&= \frac{2+\cos(\Omega)-j\sin(\Omega)}{2-\cos(\Omega)+j\sin(\Omega)} \cdot \frac{2-\cos(\Omega)-j\sin(\Omega)}{2-\cos(\Omega)-j\sin(\Omega)} \\
&= \frac{4-2\cos(\Omega)-2j\sin(\Omega)+2\cos(\Omega)-\cos^2(\Omega)-\cos(\Omega)j\sin(\Omega)-2j\sin(\Omega)+\cos(\Omega)j\sin(\Omega)\text{ -}}{4-2\cos(\Omega)-2j\sin(\Omega)-2\cos(\Omega)+\cos^2(\Omega)+\cos(\Omega)j\sin(\Omega)+2j\sin(\Omega)-\cos(\Omega)j\sin(\Omega)\text{ -}} \\
&= \frac{4-4j\sin(\Omega)-\cos^2(\Omega)-\sin(\Omega)}{4-4\cos(\Omega)+\cos^2(\Omega)+\sin(\Omega)} \\
&= \frac{4-\cos^2(\Omega)-\sin(\Omega)}{4-4\cos(\Omega)+\cos^2(\Omega)+\sin(\Omega)} + \frac{-4j\sin(\Omega)}{4-4\cos(\Omega)+\cos^2(\Omega)+\sin(\Omega)}
\end{aligned}
$$

The magnitude is found as

$$
\begin{aligned}
|H(e^{j\Omega})| &= \left| \frac{4-\cos^2(\Omega)-\sin(\Omega)}{4-4\cos(\Omega)+\cos^2(\Omega)+\sin(\Omega)} + \frac{-4j\sin(\Omega)}{4-4\cos(\Omega)+\cos^2(\Omega)+\sin(\Omega)} \right| \\
&= \sqrt{\left(\frac{4-\cos^2(\Omega)-\sin(\Omega)}{4-4\cos(\Omega)+\cos^2(\Omega)+\sin(\Omega)}\right)^2 + \left(\frac{-4j\sin(\Omega)}{4-4\cos(\Omega)+\cos^2(\Omega)+\sin(\Omega)}\right)^2}
\end{aligned}
$$

The phase is found as

$$
\begin{aligned}
\angle H(e^{j\Omega}) &= \tan^{-1}\left(\frac{\frac{-4j\sin(\Omega)}{4-4\cos(\Omega)+\cos^2(\Omega)+\sin(\Omega)}}{\frac{4-\cos^2(\Omega)-\sin(\Omega)}{4-4\cos(\Omega)+\cos^2(\Omega)+\sin(\Omega)}}\right) \\
&= \tan^{-1}\left(\frac{-4j\sin(\Omega)}{4-\cos^2(\Omega)-\sin(\Omega)}\right)
\end{aligned}
$$

Question 5 (CH 7)

(a)

The highest frequency signal between the two sinusoids is $\cos(5000\pi t)$. From this the Nyquist rate can be determined to be $\omega_s = 2 \cdot 5000\pi = 10000\pi$ rad/s. The sampling period is then $T_s = \frac{2\pi}{\omega_s} = \frac{1}{5000}$ s.

(b)

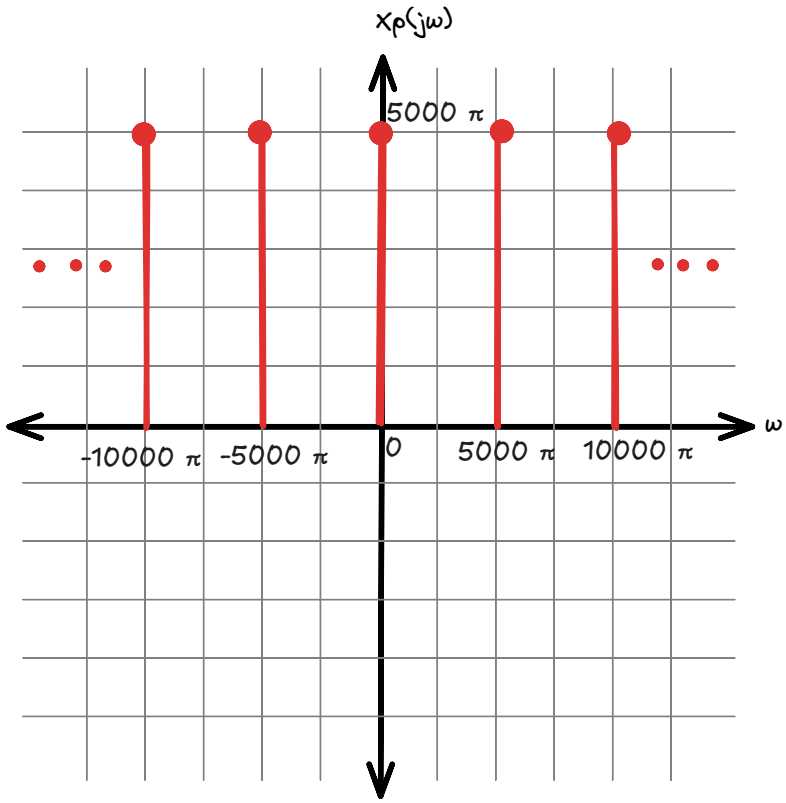
(i)

Using table 4.2 we know

$$
X(j\omega) = \pi(\delta(\omega - 5000\pi) - \delta(\omega + 5000\pi))
$$

$$
P(j\omega) = 5000\pi \cdot \sum_{k=-\infty}^{\infty} \delta(\omega - 5000\pi k)
$$

$$
\begin{aligned}
X_p(j\omega) &= \frac{1}{2\pi}(X(j\omega) * P(j\omega)) \\
&= 2500\pi \sum_{k=-\infty}^{\infty} (\delta((\omega - 5000\pi k) - 5000\pi) - \delta((\omega - 5000\pi k) + 5000\pi))
\end{aligned}
$$



(ii)

The low pass filter will filter all values in the $X_p(j\omega)$ graph out except the one at $\omega = 0$

$$
X_{out}(j\omega) = 5000\pi\delta(\omega)
$$

$$
\begin{aligned}
x_{out}(t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} 5000\pi\delta(\omega)e^{j\omega t}\, d\omega \\
&= 2500\int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t}\, d\omega \\
&= 2500e^{0jt} \\
&= 2500
\end{aligned}
$$

Since $x_{out}(t)$ is a constant function after applying the lowpass filter, it does not have any sinusoids.